Volcanism in the Earth system: volcanic eruption styles and eruption products

Part 1: Volcanic eruption styles [60 min]

The aim of this part of the practical is to interpret images of volcanic eruptions in terms of their eruption styles.

Volcanic eruption styles range from low-energy, effusive and relatively safe eruptions of lava to high-energy, explosive and hazardous eruptions of ash and gas. Eruption styles are controlled primarily by magma viscosity (i.e., whether the magma is sticky or runny) and pre-eruptive volatile content of the magma (i.e., the amount of gas, mainly H_2O and CO_2, dissolved in the magma at depth in the crust). This is because volatiles exsolve from magmas as they ascend towards the surface, producing vesicles (i.e., bubbles) that cause magmas to expand and potentially fragment (i.e., break apart) explosively.

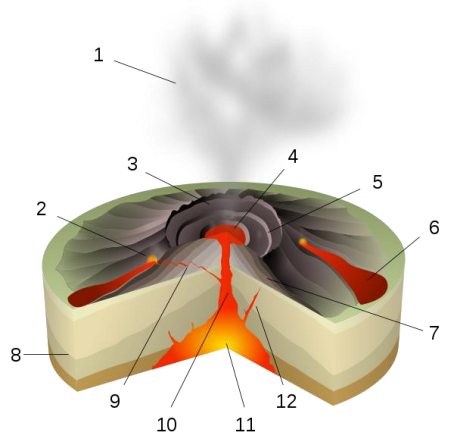
The greater the viscosity, the more magma resists expansion, and the greater the likelihood that magmas will fragment, thereby triggering an explosive eruption. Similarly, the greater the volatile content, the greater the amount that the magma expands, and therefore the greater the potential for explosive behaviour.

Magma volatile contents vary greatly, but in general magmas from mid-ocean-ridge and hotspot volcanoes have lower volatile contents than those from volcanoes in arcs at subduction zones. This is because mid-ocean-ridge and hotspot magmas are generated by decompression melting in response to mantle upwelling whereas arc magmas are generated by flux melting in response to fluid release from subducting slabs.

There are six images of different eruptions labelled B to I around the lab.

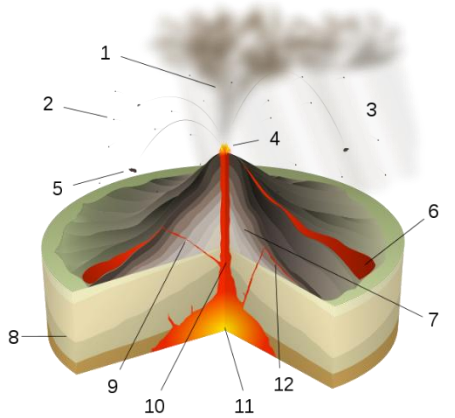
For each of the images your task is to comment on the following: [5–10 min per image]

- Is the eruption effusive or explosive? Can you observe lava, ash or both?
- Is the magma viscosity high or low (i.e., is the magma sticky or runny)?
- Is the magma likely to have had a high or low volatile content before eruption?
- Using Figures 1 and 2, what kind of eruption is the image likely to represent?
  (There may be more than one possible answer, and the style of the eruption may be unclear)
- Using Figure 3, in what tectonic setting is the magma likely to be erupting?



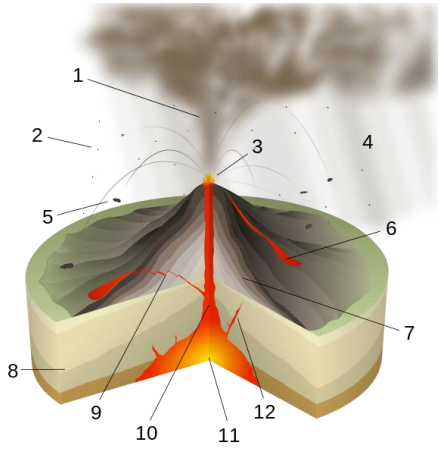
Hawaiian eruption

1. Minimal ash plume, 2. Lava fountain, 3. Crater, 4. Lava lake,
5. Fumaroles, 6. Lava flow, 7. Layers of lava and ash, 8. Country rock,
9. Sill, 10. Magma conduit, 11. Magma chamber, 12. Dyke



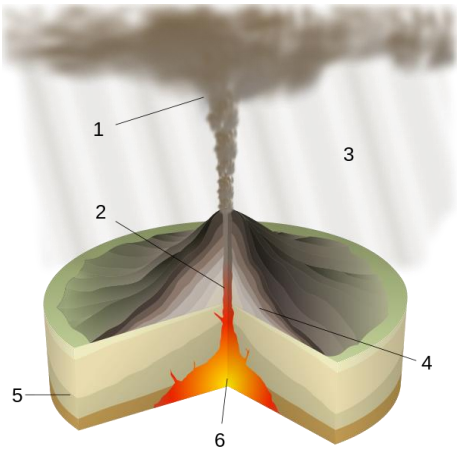
Strombolian eruption

1. Small ash plume, 2. Lapilli, 3. Volcanic ash fall, 4. Lava fountain,
5. Volcanic bomb, 6. Lava flow, 7. Layers of lava and ash, 8. Country rock,
9. Dyke, 10. Magma conduit, 11. Magma chamber, 12. Sill



Vulcanian eruption

1. Moderate ash plume, 2. Lapilli, 3. Lava fountain, 4. Volcanic ash fall,
5. Volcanic bomb, 6. Lava flow, 7. Layers of lava and ash, 8. Country rock,
9. Sill, 10. Magma conduit, 11. Magma chamber, 12. Dyke



Plinian eruption

1. Large ash plume with pyroclastic density currents, 2. Magma conduit,
3. Volcanic ash fall, 4. Layers of lava and ash, 5. Country rock,
6. Magma chamber

Figure 1 One view of volcanic eruption styles by increasing explosivity (Sémhur, CC-BY-SA-3.0).

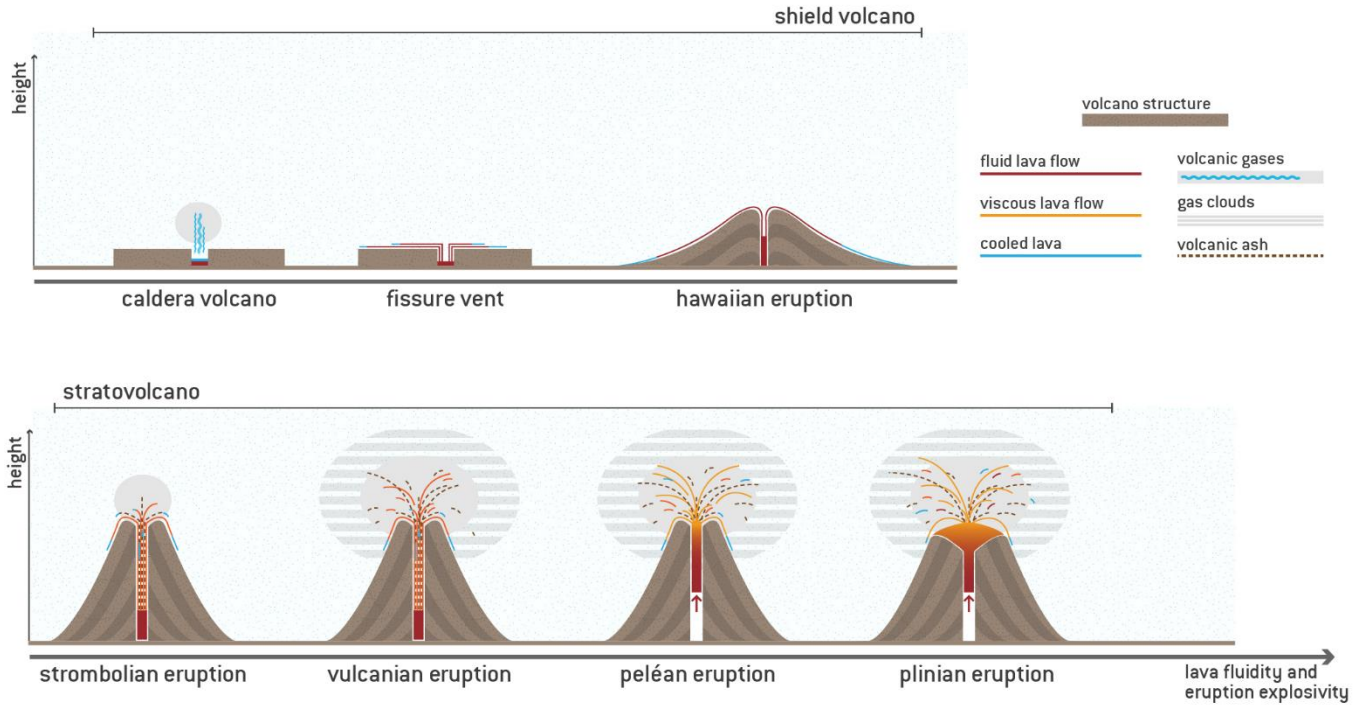


Figure 2 Another view of volcanic eruption styles by increasing explosivity (Chiara Cingottini, CC BY-SA 4.0).

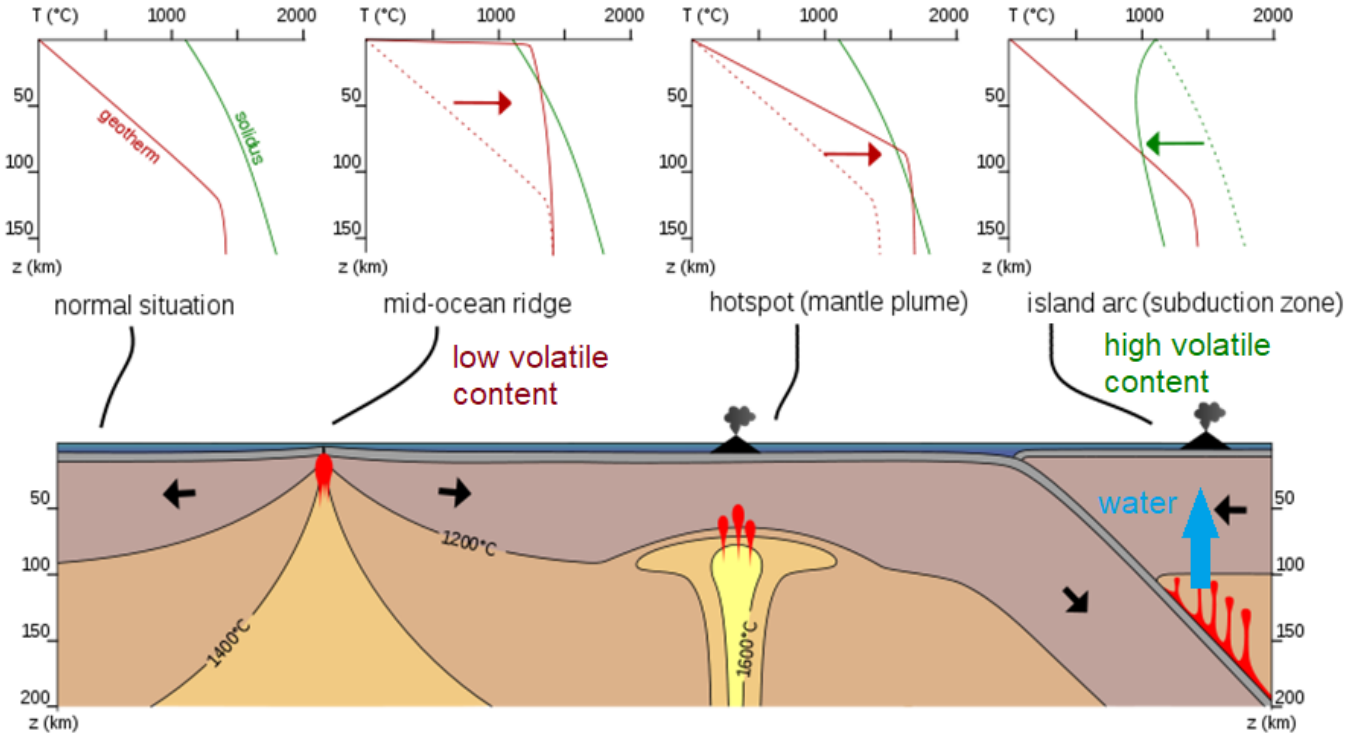


Figure 3 Magma production in different tectonic settings. Mid-ocean ridge and hotspot magmas are relatively dry while those from arcs in subduction zones are relatively wet

Part 2: Volcanic eruption products [60 min]

The aim of this part of the practical is to describe three volcanic rocks produced during eruptions of style.

The shape and texture of volcanic rocks preserve information about how they formed. For example, the density and porosity (i.e., spaces filled with fluids or gases) of volcanic rocks record information about the extent of volatile exsolution (i.e., gas release) that occurred during cooling and solidification, and therefore reflect eruption explosivity. Careful observations of volcanic rocks thus enable us to learn about past eruptions and reconstruct the history of different volcanoes. In turn, this helps us to understand and forecast the future behaviour of active volcanoes and evaluate the risks they pose to society.

There are samples of three different types of volcanic to describe. If you wish to revisit this exercise later, you can also look at 3D models of rocks that the University of Queensland, Australia, have uploaded to sketchfab.

In addition to looking at hand specimens of the samples, you can look at the basalt under the microscope to aid your description of the hand specimen and practice your optical microscopy skills.

Your task is to describe samples of *pumice*, *basalt* and *obsidian* according to the following: [20 min per sample]

- Size: What are the approximate dimensions of the sample in cm?
- Shape: Is the sample rounded, angular, elongate or blocky? Does it have the same texture all over?
- Colour: Is the sample light or dark in colour? Is it dull, shiny, or glassy? (Remember that mafic rocks (e.g., basalt) are generally (but not always) dark in colour and felsic rocks (e.g., rhyolite) light in colour)
- Mineralogy: Are there crystals in the sample? If so, how large are they? What shape and colour are they?
- Density: Is the sample light of heavy?
- Vesicularity: Are there bubbles in the sample? If so, what are their properties (size, shape, abundance).
- Eruption style: Given your answers to the above, infer a possible eruption style. In other words, did the rock form in an explosive or effusive eruption? Thinking about your answers to Part 1 may help here.

Here are some links to 3D models on sketchfab to look at outside of the practical if you wish:

- Pumice: https://sketchfab.com/3d-models/pumice-b1c57ecd8207420ba277d72a265108bd
- Basalt: https://sketchfab.com/3d-models/ropey-basalt-1ed4dc11e56745f5b01cdcfaafa6ee93
- Obsidian: https://sketchfab.com/3d-models/obsidian-972ddbb5832f41569de62748ac379bd4

If you're struggling to think about how to describe rock samples, have think about how you could describe this mystery pale brown, slightly vesicular and disc-shaped sample that is approximately 8 cm in diameter, 1 cm thick and contains brown conical 5-mm inclusions which make up about 10% of its volume...



Figure 4 Mystery sample (http://www.flickr.com/photos/87542849@N00 Kimberly Vardeman, CC-BY-2.0)

Answers and comments

Part 1: Volcanic eruption styles

Image B: Holuhraun, Iceland, 2014–2015

Volcano type: This is an effusive and long-lived eruption of lava from a small scoria cone. A fire fountain can be observed but is possibly too small for the eruption to be classified as hawaiian in style. The scoria cone could form part of an elongated fissure of similar cones. Effusive eruptions often occur on shield volcanoes or sometimes occur on stratocone volcanoes.

Eruption products: The lava has a low viscosity (i.e., is fluid), suggesting that the erupting magma is basaltic in composition. Some gas is being released from the erupting magma (though large amounts of steam can also be seen rising from the wet ground around the lava), but in insufficient quantities to supply a strong plume, suggesting that the magma is anhydrous (i.e., volatile poor).

Tectonic setting: The magma's basaltic composition and low volatile content suggest that the eruption is taking place at a mid-ocean ridge or hotspot where decompression melting in the mantle creates anhydrous basalts. Given that the eruption is subaerial (i.e., above sea level), it is probably occurring at a hotspot. Similar eruptions do nevertheless happen on the seafloor at mid-ocean ridges where they produce pillow basalts.

Image C: Sakurajima, Japan, 2013

Volcano type: This is a small and short-lived explosive eruption of ash from a steep-sided stratocone volcano. Short-lived explosive eruptions with small ash plumes (up to a few km high) are classified as vulcanian in style.

Eruption products: Despite the small size of the eruption, the magma has explosively fragmented into ash, suggesting that it is viscous (i.e., sticky), hydrous (i.e., volatile rich) and thus possibly andesitic or rhyolitic in composition.

Tectonic setting: The magma's andesitic or rhyolitic composition and its high volatile content suggest that the eruption is taking place in a volcanic arc where flux melting driven by the addition of water from subducting slabs into overlying mantle wedges creates hydrous magmas.

Image E: Kīlauea Iki, Hawaii, 1959

Volcano type: This is a mildly explosive eruption of tephra from a hidden crater. A tall fire fountain can be observed, meaning that the eruption can be classified as Hawaiian in style. Hawaiian eruptions typically occur on shield volcanoes and some stratocone volcanoes.

Eruption products: The erupting magma has broken apart into incandescent (i.e., glowing) tephra without forming a big ash plume, suggesting that it has a low viscosity (i.e., is fluid) and is basaltic in composition.

Tectonic setting: The magma's basaltic composition suggests that the eruption is taking place at a mid-ocean ridge or hotspot. Shield volcanoes are also more commonly found at hotspots than volcanic arcs.

Image F: Mt Pinatubo, Philippines, 1991

Volcano type: This is a highly explosive eruption of ash from stratocone volcano. Large explosive eruptions that generate high eruption columns (tens of km high) are typically classified as Plinian in style.

Eruption products: The erupting magma has explosively fragmented into ash, suggesting that it is highly viscous (i.e., very sticky), hydrous (i.e., volatile rich) and andesitic or rhyolitic in composition.

Tectonic setting: The magma's andesitic (or rhyolitic) composition and its high volatile content suggest that the eruption is taking place at a volcanic arc. Plinian eruptions usually occur in volcanic arcs where large volumes of silica-rich magma are created by mixing mantle and crustal melts.

Image G: Stromboli, Italy, ongoing

Volcano type: This is a short-lived and weakly explosive eruption of tephra from a small vent on a stratocone volcano. Very small eruptions of tephra are typically classified as strombolian in style.

Eruption products: The erupting magma has broken apart into incandescent (i.e., glowing) tephra, suggesting that it has a low viscosity (i.e., is fluid) and basaltic in composition. Substantial gas emissions can be observed around the erupting vent, suggesting that the magma is hydrous (i.e., rich in H_2O).

Tectonic setting: The magma's high H_2O content suggests that the eruption is taking place at a volcanic arc.

Image I: Chaiten, Chile, 2008–2010

Volcano type: This is a lava dome at the top of a stratocone volcano.

Eruption products: The lava dome has steep sides, suggesting that it is composed of very viscous (i.e., very sticky) magma that is likely to be rhyolitic in composition. Although rhyolitic magmas are typically hydrous (i.e., volatile rich), this lava dome must be degassing efficiently in order to prevent magma fragmentation from occurring. Indeed, gas can be seen escaping from the lava dome in many places. The rate at which magma is erupted must be very low to prevent explosive eruptions like those seen in C and E from happening.

Tectonic setting: The magma's rhyolitic composition suggests that it most likely formed in a volcanic arc. Rhyolites do however erupt at some hotspots.

Part 2: Volcanic eruption styles

Pumice (from sketchfab – *your exact description may vary depending on the sample you looked at*)

Size: The pumice is c. 12x10x6 cm in size

Shape: The pumice is rounded block and similar in texture on all sides

Colour: The pumice is a dull pale brown in colour, reflecting its rhyolitic composition.

Mineralogy: No crystals are visible in this pumice at the resolution of the 3D model (though crystals are found in many pumices).

Vesicularity: The sample is highly vesicular (i.e., bubbly). Visible vesicles (i.e., bubbles) range from c. 0.5 mm to 1 cm in diameter and constitute c. 50% of the pumice by volume (though in reality pumices contain 90-95 % vesicles by volume that simply are not visible at the resolution of the 3D model). Some of the vesicles are elongated.

Density: Given the high vesicularity, the pumice is likely to have a very low density.

Eruption style: The high vesicularity suggests that the pumice formed from a volatile-rich magma. Combining this high volatile content with a rhyolitic composition and the fact that the pumice forms a discrete block (i.e., a clast) indicates that the pumice formed during an explosive eruption. Indeed, pumice clasts are formed by magma fragmentation that has not proceeded sufficiently to break the erupting magma into much finer ash particles. Pumice like this could form in eruptions like those in Images C and F from Part 1.

Basalt (from sketchfab – *your exact description may vary depending on the sample you looked at*)

Size: The basalt is c. 15x15x8 cm in size.

Shape: The basalt forms a tabular block. The two largest faces have distinct textures. One has a smooth but folded appearance (i.e., has a ropey texture), while the other is rough but uniform in texture. The other four faces show a boundary between the two textures a few cm away from the smooth face.

Colour: The basalt is glassy and black on the smooth surface and grey to brown on the rough surface. The dark colours are consistent with a basaltic composition.

Mineralogy: Some small white crystals a few mm in length can be observed, but these only make up <5% of the sample volume.

Vesicularity: The smooth, glassy portion of the basalt does not contain any vesicles. The rough, grey portion of the sample contains numerous rounded vesicles c. 3 mm to 1 cm in diameter that make up c. 20-40% of the sample by volume.

Density: Given its relatively low vesicularity, the basalt is likely to be quite dense.

Eruption style: The low vesicularity suggests that the basalt formed from a volatile-poor magma. Combining this low volatile content with a basaltic composition and the fact that the basalt looks like it has been sampled from a larger body suggests that it is part of a lava flow that formed during an effective eruption. Indeed, the textures of different faces reflect that the smooth and folded surface (called pahoehoe from the Hawaiian for smooth lava) is the rapidly cooled top of a lava flow while the rough surface is part of the slowly cooled lava flow interior. Basalt like this could form in eruptions like those in Images B and E (and maybe G).

Obsidian (from sketchfab – *your exact description may vary depending on the sample you looked at*)

Size: The obsidian is c. 8x6x4 cm in size

Shape: The obsidian forms a block with very smooth faces separating by angular edges. Some of these show evidence of conchoidal fractures.

Colour: The obsidian is glassy and dark blueish grey. However, in this case the dark colour reflects the obsidian's glassy nature rather than its chemical composition; obsidians are typically rhyolitic in composition.

Mineralogy: No crystals are visible in this obsidian at the resolution of the 3D model (though crystals are found in many obsidians).

Vesicularity: No vesicles are visible in the obsidian.

Density: Given its relatively low vesicularity, the obsidian is likely to be quite dense.

Eruption style: The low viscosity of the obsidian suggests that it formed during an effusive eruption during which volatiles could escape sufficiently quickly to prevent magma fragmentation and an explosive eruption. This typically occurs when eruptions are very slow and there is plenty of time for exsolving volatiles to escape from the magma as it rises to the surface to form lava flow. In some cases obsidian can form when pumice clasts like the one described above have all of their vesicles squeezed out of them while hot, leaving only the liquid magma behind. Obsidian may form in eruptions like that in Image I.